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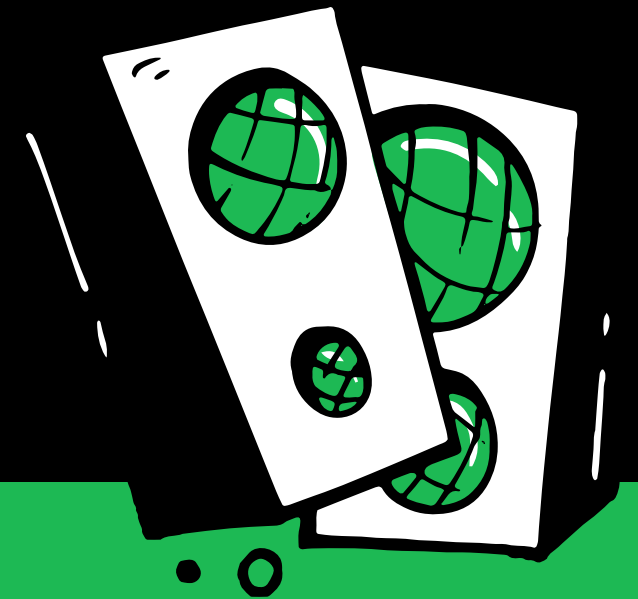
BSAN-Senior Project



What Makes a Song Popular?

A Segmented Regression Analysis of Spotify Audio Features Across
328 Artists and 25 Genres

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TEAM KESNO

Introduction



The Problem

Spotify tracks individual listening habits, not what makes a song popular within a specific artist's genre.



Our Objective

Build a linear regression model to identify which audio features drive song popularity within an artist's and their music genre(s)



Popularity Defined

Spotify's popularity score is a 0–100 index that is recency-weighted.

3 Questions Answered

- **Modeling**
 - Which audio features best predict popularity for a given artist-genre pair, and how does that vary across genres?
- **Validation**
 - Do our models explain a statistically meaningful share of popularity variance?
- **Application**
 - How can a producer, artist, or label actually use this before a song drops?



Data Collection Workflow



1

GET TOP ARTISTS

Reccobeats API used to pull the data on the top 500 artists ranked by total monthly streams

2

GET ALL TRACKS FOR EACH ARTIST

Reccobeats API: For each artist, pulled every track in their catalog, artist ID, 9 audio features, and popularity score.

3

GET GENRES FOR EACH TRACK

Matched each track to its genre using Deezer

4

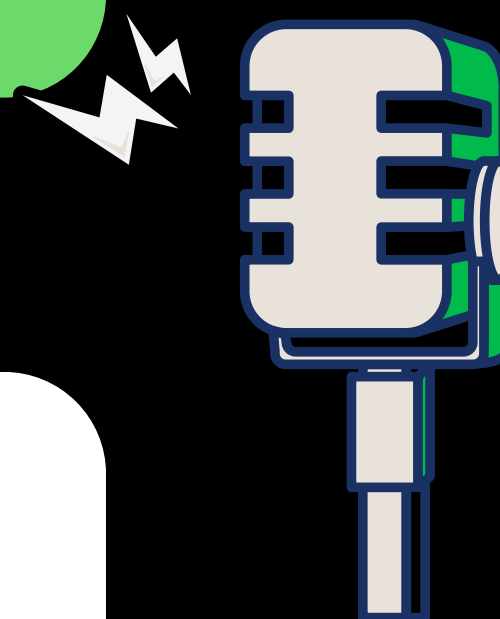
IMPUTE UNKNOWN GENRES

Used Multiple Imputation by Chained Equations to predict and fill in missing genres

5

FINAL DATA SET CREATED

Merged all sources, cleaned the data, and structured the final analysis-ready dataset.





Overview Of the Data



Train on 77,347 tracks across 25 genres to build models that actually generalize

77,347
Total
Tracks

25
Genres

328
Artists



- Acousticness
- Loudness
- Danceability
- Speechiness

AUDIO FEATURES (Predictors)

- Energy
- Tempo
- Instrumentalness
- Valence
- Liveness





Methodology



Model each artist × genre separately so predictions reflect that audience specifically

$$\text{Popularity} = \beta_0 + \beta_1(\text{Acousticness}) + \beta_2(\text{Danceability}) + \beta_3(\text{Energy}) + \beta_4(\text{Instrumentalness}) + \beta_5(\text{Liveness}) + \beta_6(\text{Loudness}) + \beta_7(\text{Speechiness}) + \beta_8(\text{Tempo}) + \beta_9(\text{Valence}) + \varepsilon$$

1

Split the data
77,347 tracks divided
into 630 artist-genre
bins



2

Run regression per bin
Separate linear
regression using 9 audio
features per bin



3

Collect outputs
 R^2 , Adjusted R^2 ,
coefficients, and p-
values per model

4

Flag significant
models
220 of 630 models hit
 $p < 0.05$



5

Compare artists
Manhattan Distance
across 19,293 artist
pairs





Why Linear Regression?

Continuous Outcome

Popularity is a 0–100 number, so regression is the natural fit.

Model Validation with R^2 and p-values

We can prove which audio features actually matter and which are just noise.

Actionable Release Decisions

A predicted popularity score for an unreleased track lets artists and labels prioritise which songs to push, when to release.

Interpretable Coefficients

Every coefficient tells us exactly how much one feature change moves the score.



Limitations We Acknowledge

It assumes linear relationships and can't capture effects like an optimal energy sweet spot.

Benchmarking Against the Genre

Knowing a song scores 72 vs. a genre average of 58 signals genuine competitive edge. ("this track is good")

Regression was chosen over **classification** (no natural threshold for 'popular') and **clustering** (no defined target variable).



★ Correlation of Features

Understanding how features relate to each other helps explain why no single formula predicts popularity across all artists.



Feature A	Feature B	r	r
Loudness	Energy	0.69	0.69
Energy	Acousticness	-0.63	0.63
Loudness	Acousticness	-0.48	0.48
Valence	Danceability	0.42	0.42
Liveness	Danceability	-0.34	0.34
Valence	Energy	0.30	0.30
Loudness	Instrumentalness	-0.25	0.25
Loudness	Danceability	0.24	0.24
Danceability	Acousticness	-0.22	0.22
Valence	Loudness	0.21	0.21





Genre Divergence Insights★

The further a genre sits from the average, the more its audience demands a unique production strategy.

Brazilian Music (0.182)

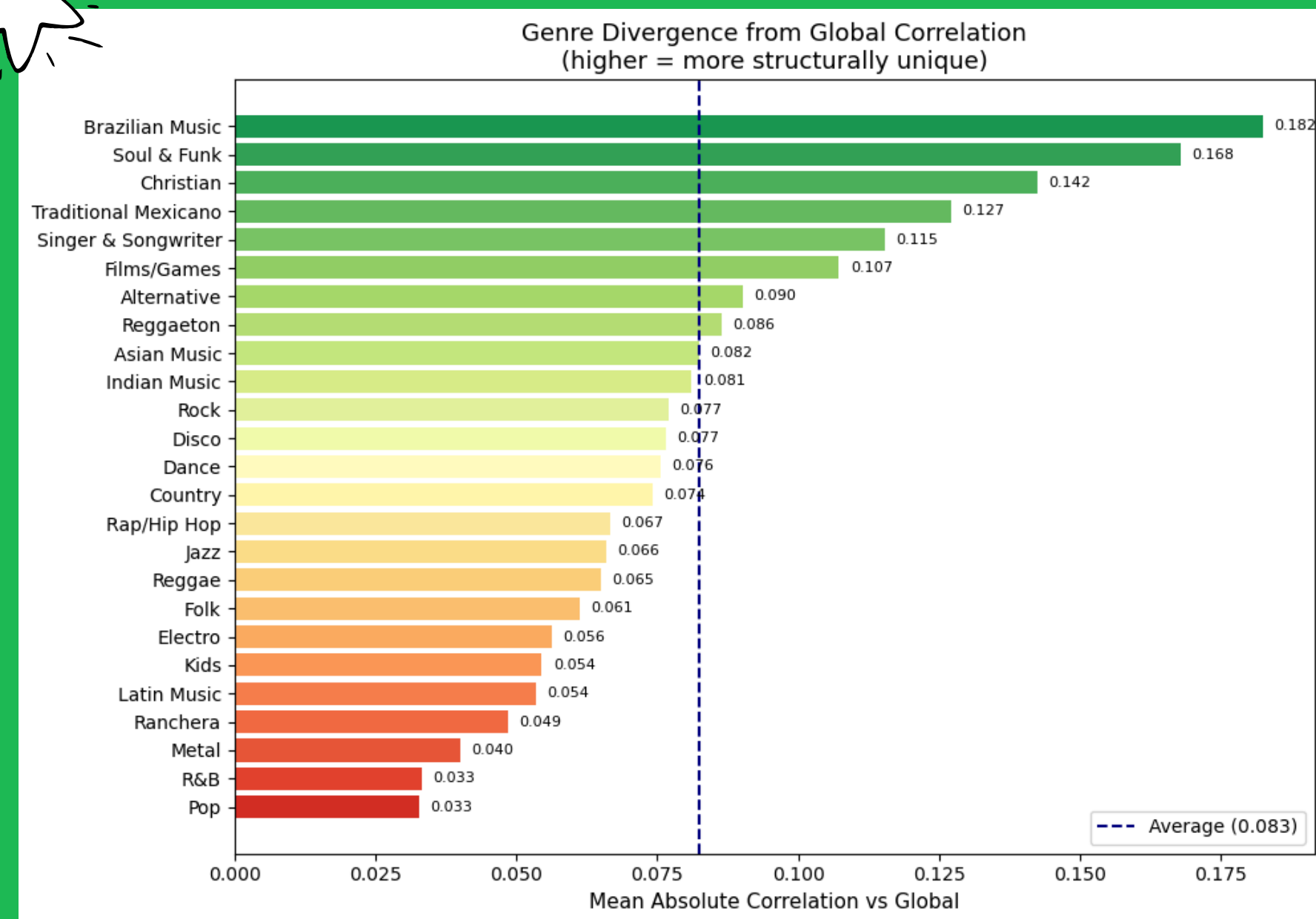
Brazilian Music diverges the most from the global average, meaning audio features like energy and danceability predict popularity very differently here than they do platform-wide

Reggaeton (0.086)

Reggaeton sits right at the global average, meaning its feature-to-popularity relationships are fairly consistent with broader Spotify trends

Pop (0.033)

Pop has the lowest divergence of any genre, meaning its audio feature relationships are the most predictable and closest to the global norm, making it the most straightforward genre to model and optimize for.



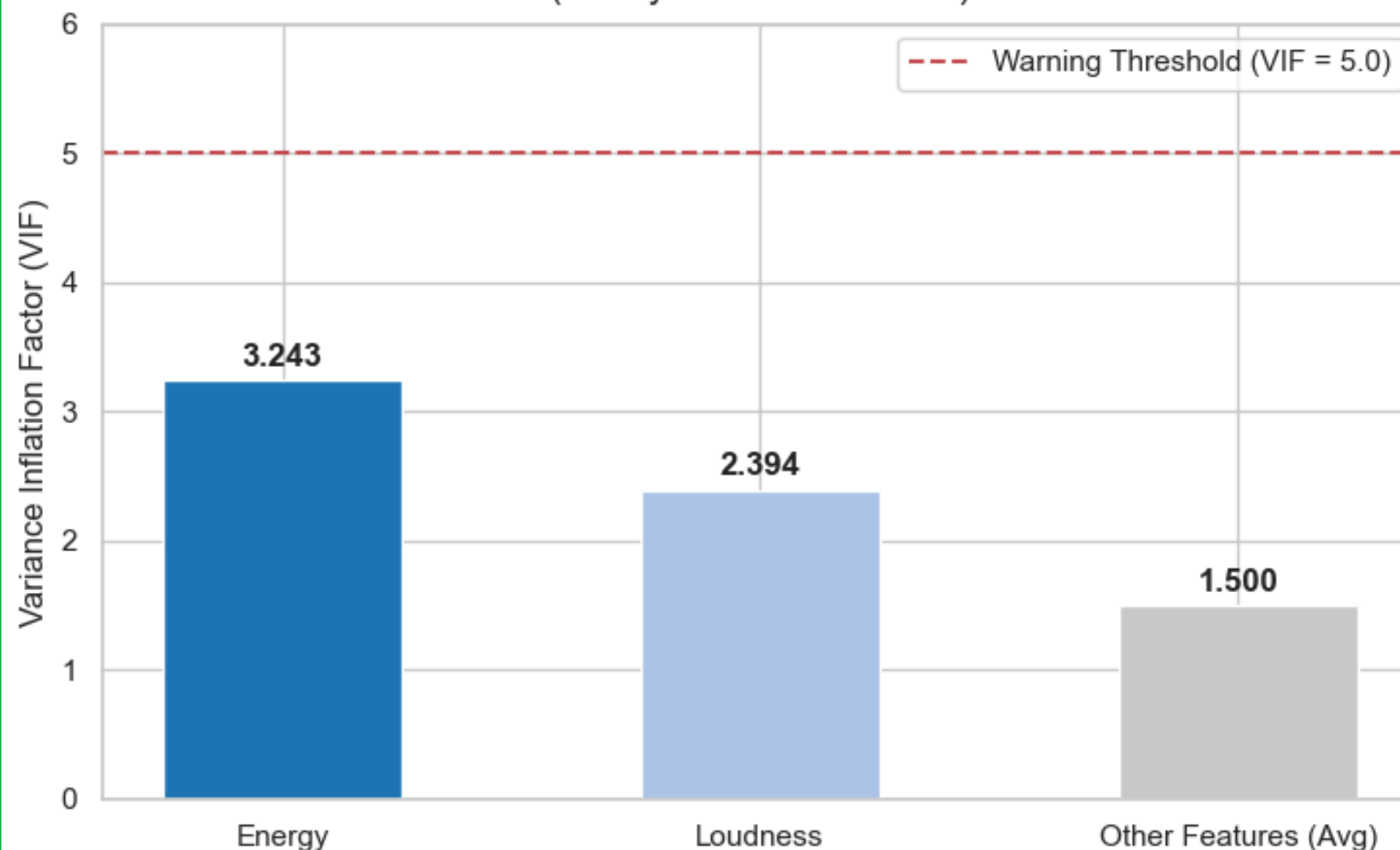


Multicollinearity Test

VIF - Is our model overfitting due to the feature correlations?

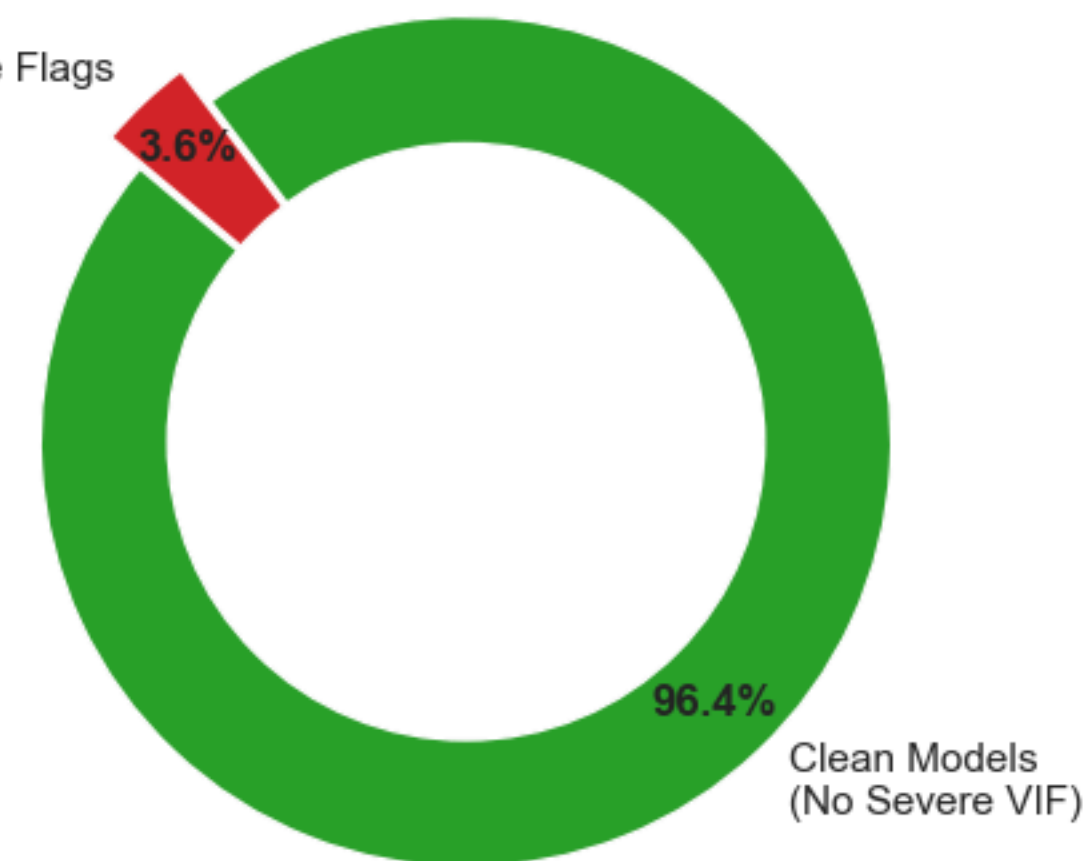


Global Feature VIF Summary
(Safely Below Threshold)



Model Cleanliness Rate
(220 Total Significant Models)

Models w/ Severe Flags





Sample Size of Genres



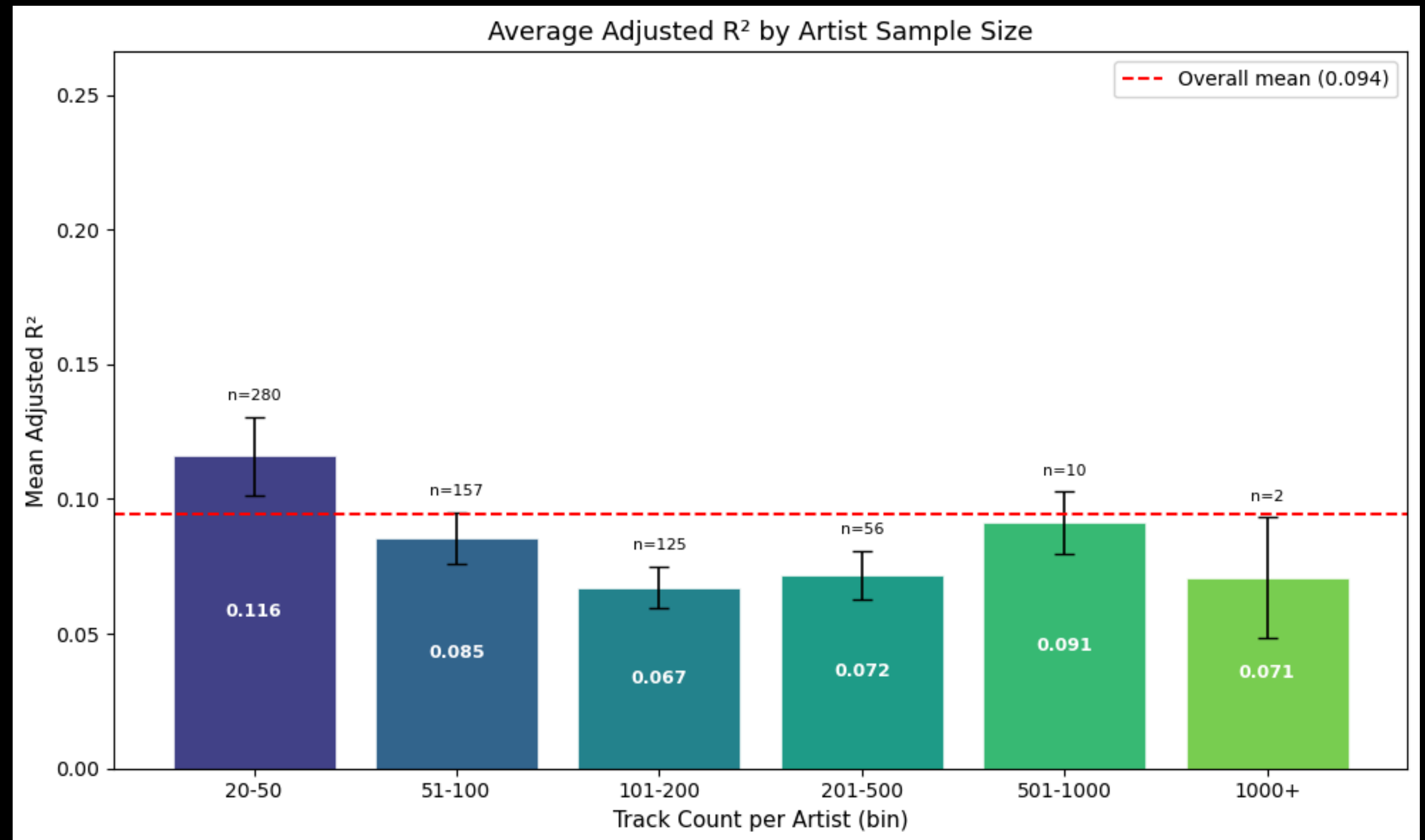
Sample sizes vary across each artist-genre pair - is the model predictive across those sizes?

20-50 Track Bin

Higher adjusted R^2 in this first bin than any others

50-1000+ Track Bins

Adjusted R^2 is fairly similar across the board between the rest of the bins





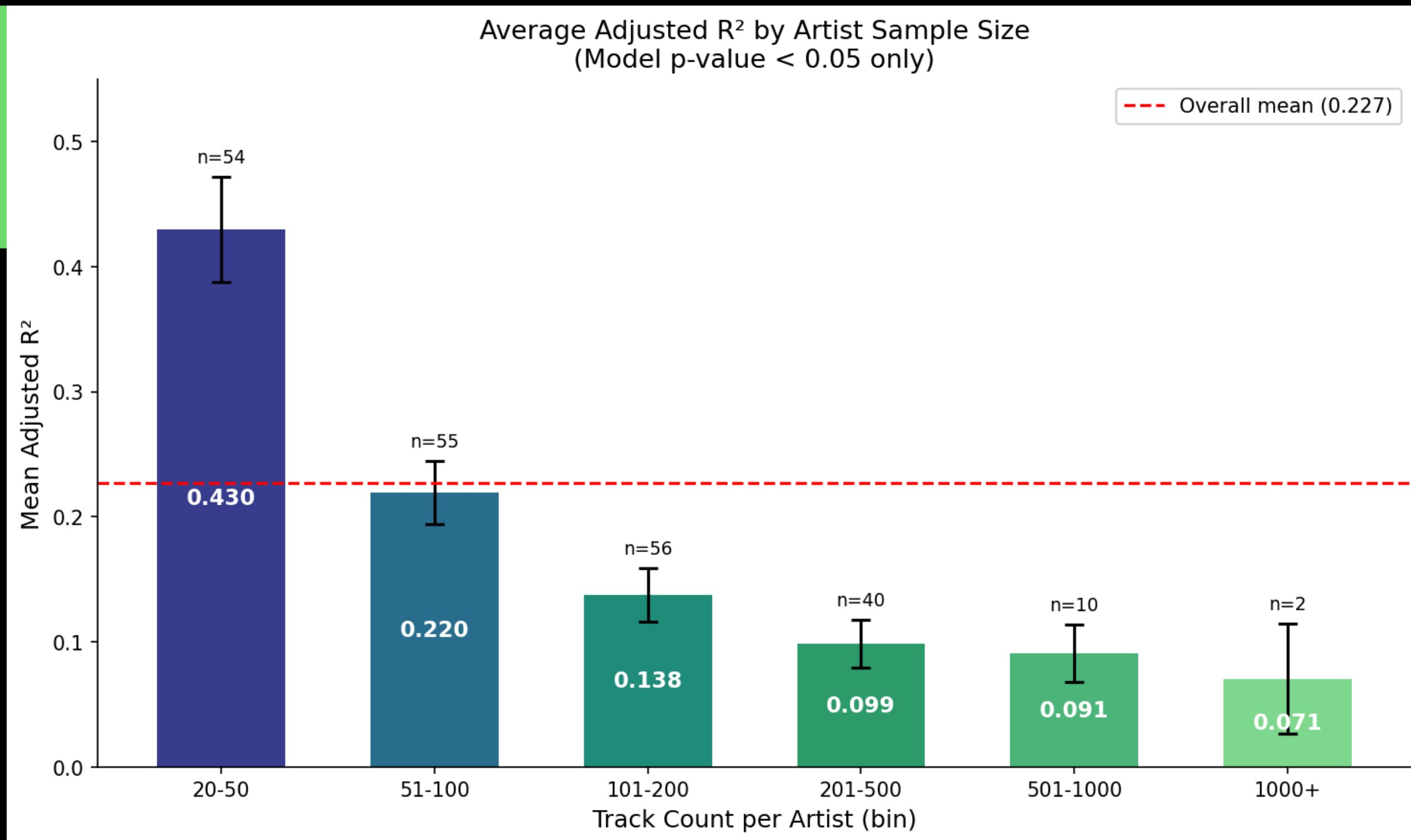
Narrowing to Significant Models



Sample sizes vary across each artist-genre pair - is the model predictive across those sizes?

What about when we only take the statistically significant models? (p-value < 0.05)

- This shows that the model focuses more on what can make a song more popular than actually predicting the popularity
- For the non-significant models, outside factors like marketing, placement, and timing could affect the popularity



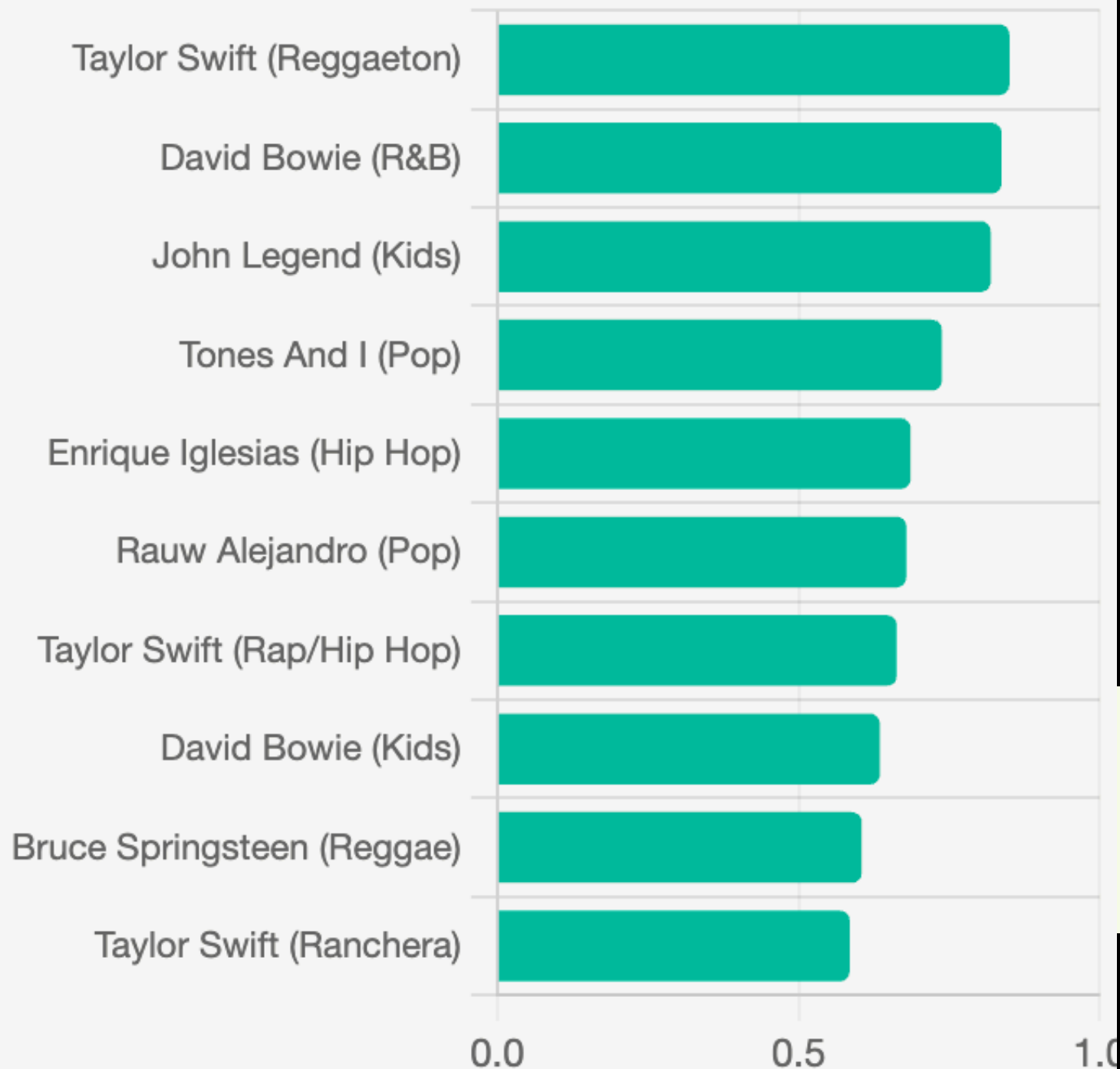


Model Performance



Does this model accurately predict song popularity from the given features?

Top artists by adjusted R²



Example

Taylor Swift (Reggaeton)

This model explains 85% of the variation in popularity based on the 9 audio features we used

The model is statistically significant, with a model p-value of 0.0001.

Popularity Equation

$$\text{Popularity} = 30.5238 + -5.1669(\text{Acousticness}) + 5.4025(\text{Danceability}) + 16.206(\text{Energy}) + -4.732(\text{Instrumentalness}) + -7.1987(\text{Liveness}) + -0.9128(\text{Loudness}) + -3.1184(\text{Speechiness}) + 5.2005(\text{Tempo}) + 1.9458(\text{Valence})$$

Model Results

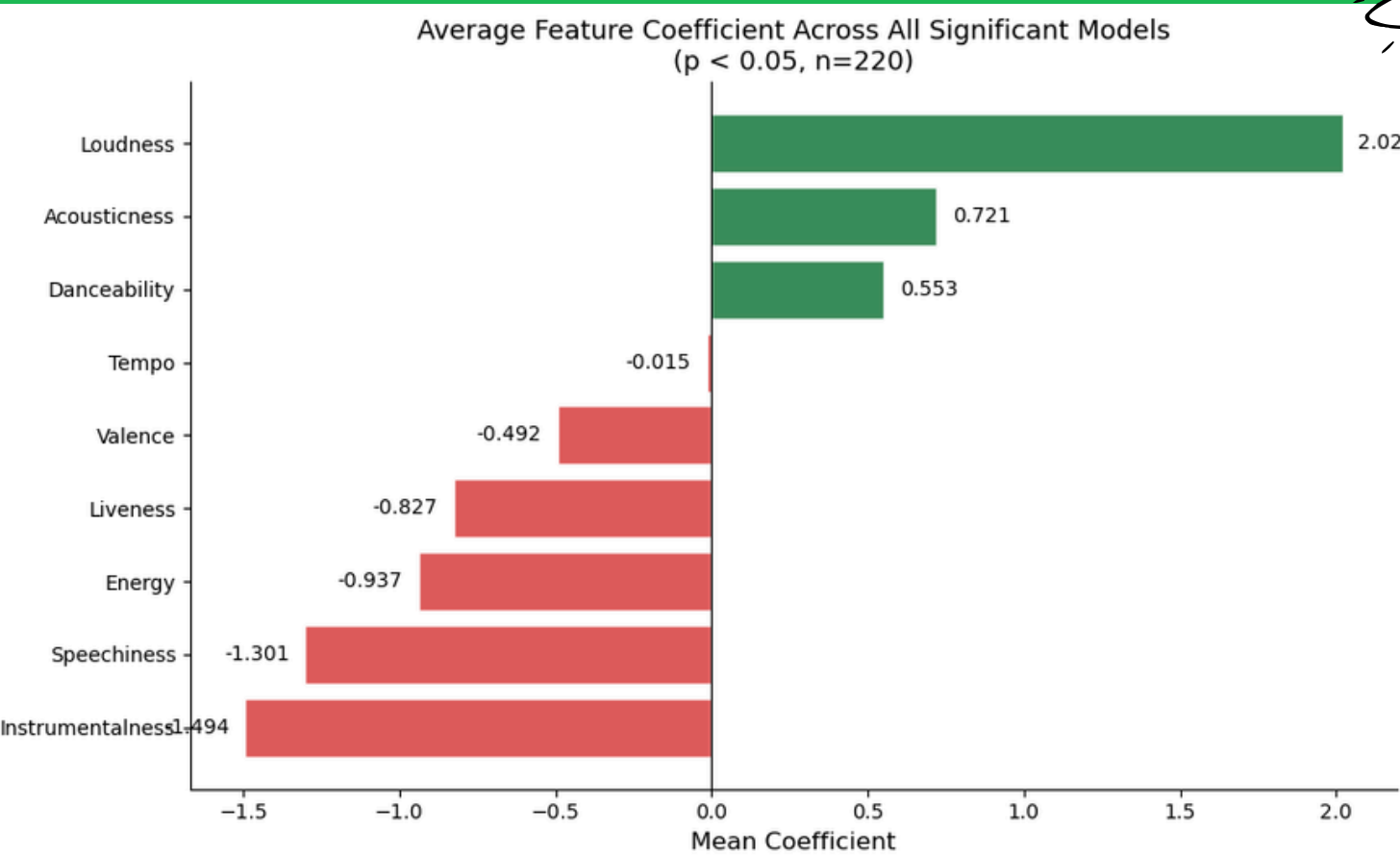
What were the findings from the audio features?

Dominant Audio Features

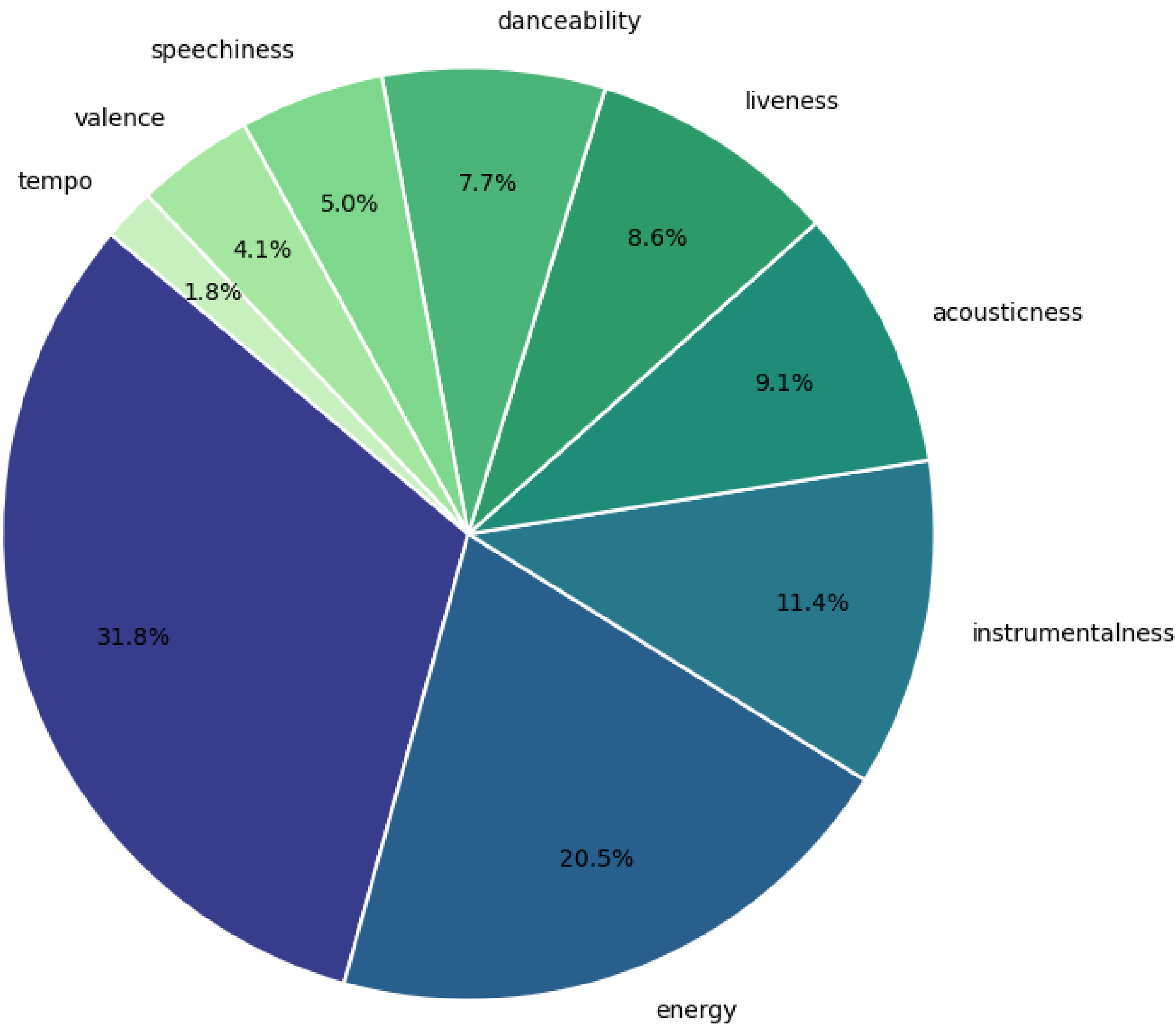
The most dominant features are loudness, energy, and instrumentalness
Loudness and energy also have a strong correlation

Feature Affects

Loudness, acousticness, and danceability have a positive Affect on popularity, tempo is about even, and others are negative



Most Dominant Feature by Artist
(Significant Models, $p < 0.05$, $n=220$)





Manhattan Distance Results



We calculated the total difference in audio feature coefficients between every artist pair in the same genre:
The smaller the number, the more alike their sound.

David Bowie vs The Beatles

Cross-Genre Comparisons

Genre

Distance

Pop

10.0

Rock

16.9

Kids

39.7

Elvis Presley vs Taylor Swift

Cross-Genre Comparisons

Genre

Distance

Pop

64.0

Ranchera

58.2

Indian Music

43.6

Labels can use distance scores to identify crossover opportunities, avoid mismatched collaborations,
and target the right audience when pitching a new track.
Similarity = closest production blueprint for popularity in their shared genre — not that they sound alike.

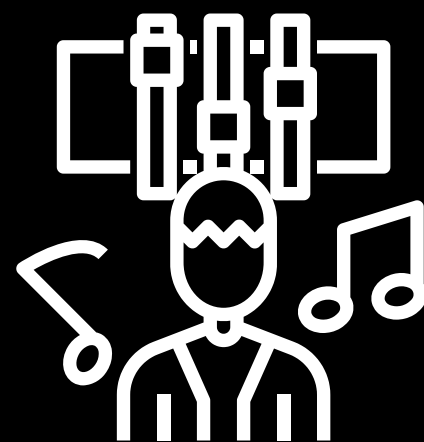


Business Implications Example



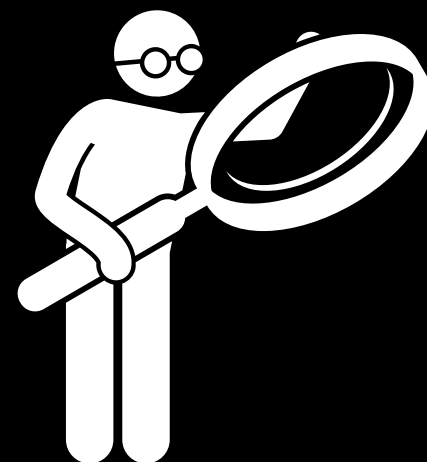
Enrique Iglesias is releasing a new Hip Hop single

STEP 1



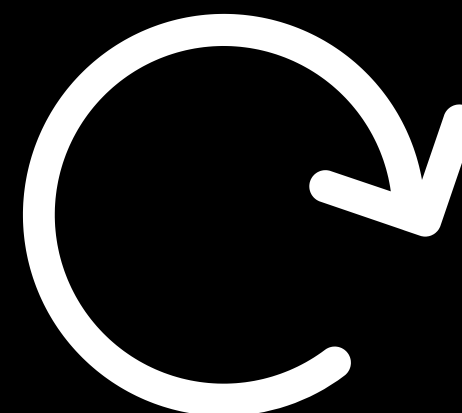
Produce
Mix
Master

STEP 2



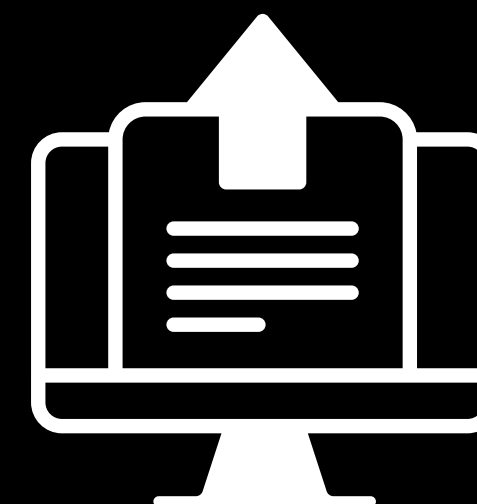
Analyze Audio
Features

STEP 3

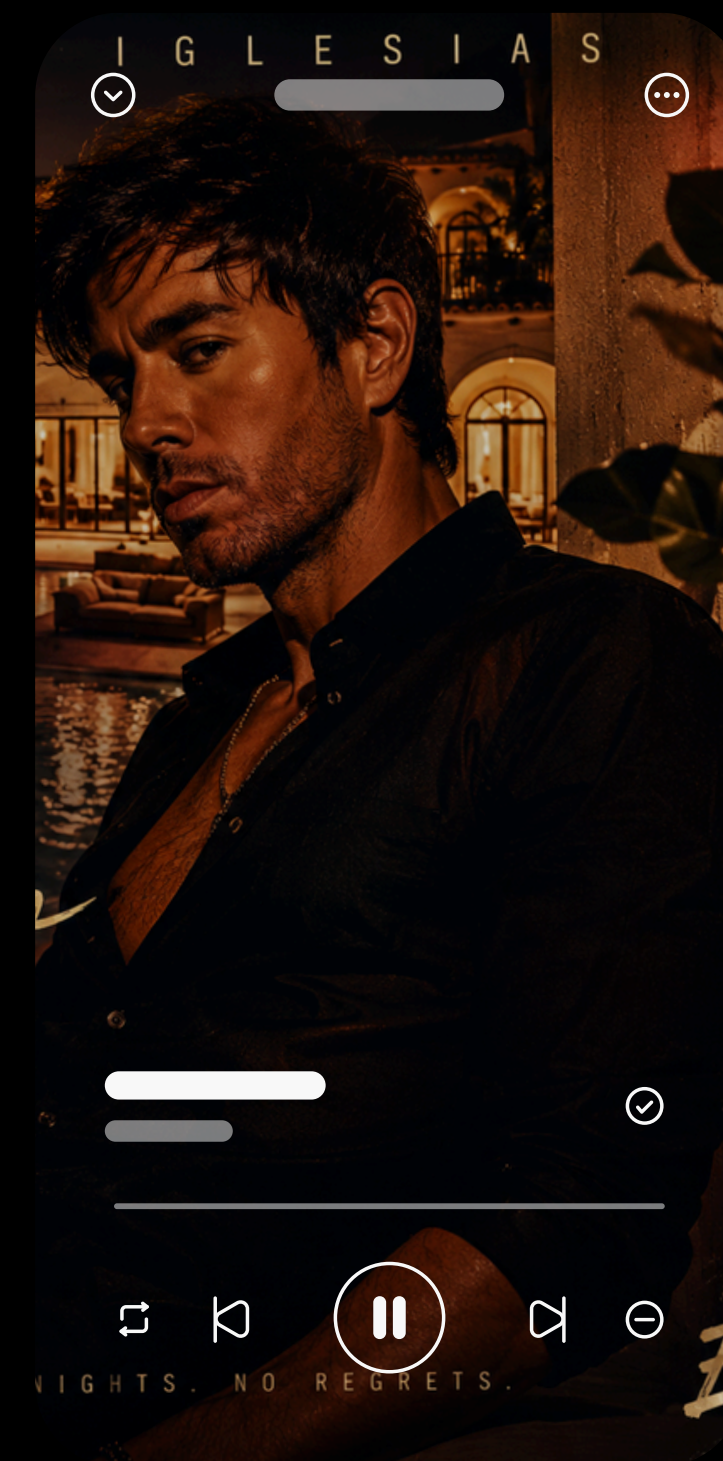


Change track
composition

STEP 4



Publish Song





Business Implications



TARGET CUSTOMER SEGMENTS



01

Record Labels

Use our model to make data-backed A&R decisions, score tracks before release, and match artists using Manhattan distance.



02

Independent Artists

Get feature-level feedback on their tracks vs. the top 328 artists and spot genre mismatches before releasing.



03

Creating & Owning a Record Label

Every signing gets a chart readiness score. First-mover advantage in ML-driven indie A&R with a built-in SaaS revenue stream.

EACH SEGMENT EVALUATED ACROSS



Model Benefits



Strengths



Weaknesses



Opportunities



Threats



Business Implications

Target Customer: Record Labels

MODEL BENEFITS

- Score tracks before release
- Match artists via Manhattan distance
- Genre-specific production strategy

STRENGTHS

- 630 genre-bin models, highly granular
- Statistically validated ($p < 0.05$)
- Reduces costly A&R guesswork

WEAKNESSES

- Low avg R^2 (~0.10–0.25)
- Audio features alone don't explain all popularity

OPPORTUNITIES

- License as A&R screening tool
- White-label API for label dashboards
- Expand to playlist placement prediction

THREATS

- DSPs build competing internal tools
- Bad prediction damages credibility
- Labels build in-house data teams



Business Implications

Target Customer: Independent Artists

MODEL BENEFITS

- Feature-level feedback vs. top 328 artists
- Spot genre mismatches before release
- Actionable mixing guidance from coefficients

STRENGTHS

- No consumer tool offers this granularity
- 25-genre coverage incl. niche markets
- Artists can iterate mix using outputs

WEAKNESSES

- Coefficients hard to interpret without training
- Dataset skews toward already-popular artists
- Recency-weighting may bias older training tracks

OPPORTUNITIES

- \$19–29/mo SaaS subscription model
- Bundle with DistroKid / TuneCore

THREATS

- Artists game the model, lose authenticity
- Free AI tools undercut our pricing
- Poor early results kill word-of-mouth



Business Implications

Target Customer: Creating & Owning a Record Label

MODEL BENEFITS

- Every signing gets a chart readiness score
- Target genres with high R^2 & low competition
- Distance scores guide feature pairings

STRENGTHS

- We built it — full control & interpretability
- Model improves as our roster grows
- No indie label uses ML A&R at this level

WEAKNESSES

- 65% of bins not statistically significant
- Can't predict virality or social momentum
- Limited capital to act on insights

OPPORTUNITIES

- Focus on high- R^2 genres (Reggae, Trad. Mexicano)
- License model to other small labels
- First-mover in ML-driven indie A&R

THREATS

- Methodology easily replicated by competitors
- Spotify API restrictions block retraining
- Over-reliance misses culturally viral artists



KEY TAKEAWAYS

- Audio features predict popularity differently for every artist and genre
- Loudness, energy, and instrumentalness are the strongest drivers of popularity across the board
- Pop is the most predictable genre to optimize for; Brazilian Music is the least
- Knowing your artist's audio fingerprint before release reduces costly guesswork
- Manhattan Distance tells you who your artist sounds like — useful for collaborations, A&R, and audience targeting
- This model gives you a competitive edge — no tool on the market offers artist-genre level prediction at this scale



Appendix



1. Scope of Data

- 77,347 tracks
- 328 artists
- 32 unique genres
- 21.53 average popularity

2. Correlation Matrix

- Strongest synergy was between loudness and energy, with a 0.69 positive correlation
- These relationships diverge across genres, with an average of 0.083 off the total correlation matrix

3. Model Performance

- 630 models run
- 220 statistically significant models
- Adjusted R^2 ranges from 0.85 to 0.0204 for significant models

4. Dominant Audio Drivers

- Loudness is most common at 165 models (26.2% total), and for significant models, 70 models (31.8%)
- Loudness, Energy, and danceability are, on average, positively correlated to popularity

5. Cross-Artist Feature Pairing

- 2,945 unique across-artist genre pairings calculated
- On average, 32.82 total average distance apart from every artist



QUESTIONS?



Thank you

